

Phuong-Linh Huynh-Ha

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RESEARCH EXPERIENCE

- **KIT Computer Vision Lab – Kyoto Institute of Technology (KIT)** Jan. 2026 – Mar. 2026
Research Intern, supervised by Prof. Nobuhara Shohei
 - Selected for the Global Internship Program of KIT and granted the JASSO Scholarship from Japanese government for academic excellence.
 - Conducting research on **3D reconstruction from video** with near-rotational motion and low parallax, leveraging **2D correspondences** to resolve motion ambiguity in **low-texture, non-distinct environments**.
- **Software Engineering Laboratory – SELab HCMUS** Aug. 2025 – Present
Thesis - Supervised by Assoc. Prof Minh-Triet Tran on Computer Vision
 - Researching on surgical scene reconstruction.

SELECTED PUBLICATIONS

- Hoang-Phuc Nguyen, **Phuong-Linh Huynh-Ha**, Minh-Triet Tran. **Generalizability Evaluation and Anchor-Guided Approach for Category-Agnostic Pose Estimation**. SOICT 2025, Springer CCIS.

EDUCATION

- **University of Science (HCMUS), Ho Chi Minh City, Vietnam** Sep. 2022 – Present
Advanced Program in Computer Science; GPA: 3.92/4.0
 - Granted **Merit-Based Scholarship (Top 4 GPA)** in Semester 2, 2024-2025.
 - Being 1 of **5 recipients** of **Career Readiness Scholarship**, awarded by **Gameloft** in partnership with **HCMUS** for demonstrating strong professional potential and academic merit.
 - Achieved **Top 1.9% score on the VNU-HCMC Assessment Test (HSA)**
 - Relevant coursework: Artificial Intelligence, Computer Vision, Natural Language Processing, Information Retrieval, Computer Systems, Statistics.
- **Le Khiet High School for the Gifted, Quang Ngai, Vietnam** Sep. 2019 – Jun. 2022
Major in English; GPA: 9.7/10.0
 - **Bronze Medal** in the National Olympiad in English, held by Vietnam Ministry of Education and Training.
 - **Consistently ranked Top 1 GPA** throughout all years of the gifted English student program.
 - Granted **Odon Vallet Scholarship** for academic excellence, issued by Rencontres du Vietnam.

SELECTED ACHIEVEMENTS AND AWARDS

- **Artificial Intelligence Olympiad for University Students (3-person team)** Nov. 2025 – Dec. 2025
Organized by Vietnam Association for Information Processing (VAIP)
 - Awarded **Third Prize** at the **National Finals** (Dec. 2025).
 - Awarded **First Prize, Champion** across **all Regional Preliminary Rounds** (Nov. 2025).

SELECTED PROJECTS

- **AI-Powered Media Intelligence Platform**
 - Built an AI media platform (**Django, Elasticsearch, Celery-Redis**) for media retrieval using natural language.
 - Engineered a scalable **RAG pipeline (LLM, STT, TTS)** and a controller-worker architecture to power a conversational **3D AI assistant**.
- **EndoFGS: Surgical Scene Reconstruction & 3D Gaussian Splatting** Researching
 - Enhancing endoscopic surgical scene reconstruction by leveraging high/low-frequency data to enrich feature vectors.
- **SoccerNet SynLoc: Athlete Detection & Localization** Researching

- Optimizing monocular athlete detection via synthetic data, calibration, and low-resolution object detection.
 - Implementing **Pose Estimation** for pelvis keypoint extraction, improving 2D-to-3D projection from pixels to stadium coordinates.
- **Semantic News Retrieval System**
 - Architected a semantic search engine using **OpenAI's CLIP** to embed video keyframes and text for natural language visual search.
 - Built a scalable vector search backend (**Milvus, Python**) to index and query high-dimensional embeddings with latency within a few seconds.
- **Historical Character Recognition (Sino-Nom)**
 - Distinguished easily confused characters by designing an **evolutionary algorithm-based dictionary**, using custom loss metrics and voting to iteratively optimize the candidate population.
 - Generated richer **feature vectors** by extracting and analyzing smaller critical sub-components of characters, significantly improving model accuracy on visually similar data.
- **Lightweight Image Classification**
 - Developed lightweight hybrid CNN-attention models (**EdgeNeXt, 1.1M params**) achieving **83.22% top-1 accuracy**.
 - Improved convergence and robustness using a custom **StableSAM** optimizer, advanced schedulers, label smoothing, and augmentations (**AutoAugment, MixUp, CutMix**).
- **Smart-Note-Assistant** Handled UI & API part
 - Worked in a team, engineered a responsive **Kotlin** UI for note-taking with non-blocking, **asynchronous API connections** for real-time synchronization.
 - Implemented efficient **data serialization** workflows to manage complex objects, ensuring high performance and application data integrity.

EXTRACURRICULAR ACTIVITIES

- **Youth Union Branch Commendation (Youth Union Branch 22APCS2)** 2022–2023
Committee Member – University of Science, VNU-HCM
 - Received a **Certificate of Merit** from the Principal for achieving good results in Youth Union organizational work and youth movements during the 2022-2023 academic year.
- **Inspire to Aspire Seminar Series** 2023
Organizer – Ho Chi Minh City, Vietnam
 - Organized a seminar series titled “*Inspire to Aspire*” that connected students with experienced professionals in technology and business, providing career insights and fostering industry awareness.
- **Asian-Pacific Children's Convention (APCC)** Jul. 2015
Youth Delegate – Fukuoka, Japan
 - Selected by the Ho Chi Minh Communist Youth Union to represent Vietnam at APCC in Fukuoka, Japan, focusing on international cultural exchange to foster global understanding, peace, and friendship.

SKILLS & INTERESTS

Technical Skills:	Scientific Research, Problem Solving, Machine Learning, Computer Vision
Technical Languages:	C/C++, Python, Kotlin, SQL, LaTeX, Markdown, CSS, HTML
Frameworks/Libraries:	PyTorch, NumPy, Pandas, R, Jetpack Compose, SFML
Tools:	Google Colab, Docker, Git/GitHub, Android SDK, 3D Slicer
Languages:	English (IELTS 8.0), Vietnamese (native), Japanese (for daily conversation)
Interests:	Cycling, Swimming, Singing, Guitar (learning), Poker